

ArTE-VLA: Autoregressive Target Effects for Vision-Language-Action Models

Anonymous Authors

Abstract—Vision-language-action (VLA) models typically generate robot actions directly from visual observations and language instructions, leaving the intended physical evolution of the scene implicit. We introduce ArTE-VLA, a VLA that predicts how the language-referred target should change before generating an action. Given dual-view RGB observations and an instruction, ArTE-VLA infers an implicit target representation and autoregressively predicts target effects at four future horizons. Each effect describes translation, rotation, linear and angular velocity, and activity. The predicted effects are embedded as tokens and appended to the visual-language prefix, allowing the action expert to condition on a compact physical future. Simulator states provide effect supervision during training but are not required at inference. In a single-seed study on all ten LIBERO-Object tasks, ArTE-VLA improves the success rate of a recipe-matched PI0.5 baseline from 70.8% to 83.6% over 500 paired episodes. Fixed-noise interventions further show that changing the predicted effect values produces consistent changes in generated actions. These results suggest that a compact, target-centric future representation can provide an effective and inspectable interface between VLA perception and control.

TODO: Recompute the abstract’s numerical claims after multi-seed replication. Until then, retain the explicit single-seed qualification.

I. INTRODUCTION

Vision-language-action models have emerged as a promising route toward general robot policies. By combining pre-trained visual-language representations with large collections of robot demonstrations, recent VLAs can follow open-vocabulary instructions and transfer across objects, tasks, and environments [1]. Most existing architectures, however, are optimized to map the current observation directly to an action sequence. The physical outcome that connects an instruction to an action—which object should move, in what way, and over what time scale—remains latent inside the policy.

This direct formulation creates two related difficulties. First, the action expert must simultaneously resolve language grounding, infer task-relevant dynamics, and generate precise controls. Second, the resulting computation is difficult to inspect: a successful rollout does not reveal whether the policy anticipated the intended object motion or relied on a less robust correlation. Predictive world models and visual chain-of-thought methods address this problem by forecasting future images or point trajectories before acting [3], [4], [6], [7]. Although expressive, these intermediate representations can be high-dimensional and need not isolate the physical state of the object that is central to the instruction.

We propose ArTE-VLA, which inserts a compact target-effect bottleneck into a pretrained VLA. Rather than predicting a complete future image, the model compresses its visual-language prefix into an implicit representation of the

primary target and predicts the target’s physical effects at four horizons. Each prediction contains a 3-D translation, a 3-D rotation, linear and angular velocity, and an activity logit. The predictions form an autoregressive chain: the effect at one horizon is used to predict the next. Horizon-specific hidden states and effect values are then projected into tokens that condition the flow-matching action expert.

The target-effect representation is learned with privileged supervision but used without privileged inputs. We recover object states from the simulator states stored in LIBERO demonstrations and construct aligned multi-horizon effect labels. These labels appear only in the auxiliary prediction loss; the action expert always receives model-predicted effects. At test time, ArTE-VLA uses the same RGB observations, instruction, and robot state as vanilla PI0.5 . This separation allows structured physical supervision to shape the policy without introducing an object detector, simulator state, task identifier, or ground-truth target state into the deployed model.

Our main contributions are:

- We introduce a compact, target-centric physical-effect bottleneck for VLA policies. The bottleneck predicts a four-horizon autoregressive future and directly conditions the action expert.
- We develop a replay-based labeling pipeline that adds aligned target effects to robot demonstrations while preserving their original images, actions, states, and language instructions.
- We formulate value-preserving interventions that separate reliance on the complete effect token from reliance on its explicit physical values and temporal order.
- On LIBERO-Object, ArTE-VLA improves a matched PI0.5 baseline by 12.8 percentage points in a 500-episode paired evaluation, while revealing substantial variation across target objects.

II. RELATED WORK

A. Vision-Language-Action Models

VLA models adapt pretrained vision-language representations to continuous robot control. PI0.5 combines heterogeneous co-training with a flow-matching action expert to support open-world manipulation [1]. ArTE-VLA uses PI0.5 as its backbone and leaves the original visual-language prefix and action head intact. Its contribution is an intermediate physical prediction interface that is trained jointly with the action objective and consumed directly by the action expert.

B. Predictive Representations for Control

World-model approaches learn scene dynamics through future observation prediction. GR-2 jointly learns video generation and action prediction from web videos and robot trajectories [5], while WorldVLA unifies image and action generation within an autoregressive model [4]. CoT-VLA generates future visual goals as an explicit visual chain of thought before predicting actions [3]. These methods represent future scene content with image or video tokens. In contrast, ArTE-VLA predicts a low-dimensional physical state change for one language-conditioned target. The two approaches are complementary: visual prediction preserves scene appearance, whereas a target-effect bottleneck exposes a concise variable that can be measured and intervened upon.

C. Object and Point Motion as Policy Guidance

Predicted motion can also serve as a control representation. Any-point Trajectory Modeling predicts the future motion of arbitrary image points and uses those trajectories to guide policy learning [6]. Track2Act predicts goal-conditioned point tracks from internet videos and converts them into object and end-effector motion plans [7]. ArTE-VLA shares the premise that future motion is useful for control, but differs in three respects: it predicts a single instruction-grounded target, represents motion with multi-horizon 3-D physical effects rather than 2-D tracks, and injects the prediction into the token sequence of an end-to-end VLA action expert.

TODO: Expand and update the related-work review before submission, including target-conditioned object representations, action-conditioned world models, and contemporaneous VLA methods. Verify the final publication metadata for every arXiv citation.

III. METHODOLOGY

A. Problem Formulation

At time t , the policy receives RGB observations I_t from a workspace and a wrist camera, robot state s_t , and language instruction ℓ . It predicts an action chunk $A_t = (a_t, \dots, a_{t+C-1})$. We augment this mapping with a target effect chain $E_t = (e_t^1, \dots, e_t^H)$ at $H = 4$ future horizons:

$$p(E_t | I_t, s_t, \ell) = \prod_{h=1}^H p(e_t^h | e_t^{<h}, I_t, s_t, \ell), \quad (1)$$

$$p(A_t | I_t, s_t, \ell) = p_\theta(A_t | X_t, T(E_t)), \quad (2)$$

where X_t denotes the original PI0.5 prefix and $T(E_t)$ denotes the predicted effect tokens. We use frame offsets (1, 4, 8, 16), corresponding to (0.1, 0.4, 0.8, 1.6) seconds at the 10-Hz control rate. Autoregression in Eq. (2) refers to temporal prediction and does not imply identification of a structural causal model.

B. Implicit Target Representation

PI0.5 encodes the two RGB observations and instruction into prefix embeddings $X_t \in \mathbb{R}^{N \times d}$. We apply masked mean

pooling over the prefix and project the result to a 20-D vector:

$$z_t = f_{\text{target}}\left(\frac{\sum_{i=1}^N m_i X_{t,i}}{\sum_{i=1}^N m_i}\right) \in \mathbb{R}^{20}. \quad (3)$$

The vector z_t is the sole active entity input to the effect predictor. It is not directly regressed to the current simulator state; instead, its semantics are induced by language-conditioned future-effect supervision and the action loss. We therefore treat it as an implicit target representation rather than a calibrated estimate of object pose.

C. Multi-Horizon Effect Chain

The effect model projects z_t into a hidden state and combines it with a pooled VLA context, a learned horizon embedding, and the previously predicted effect. At horizon h , it produces

$$e_t^h = [\Delta p_t^h, \Delta r_t^h, v_t^h, \omega_t^h, q_t^h] \in \mathbb{R}^{13}, \quad (4)$$

where $\Delta p_t^h \in \mathbb{R}^3$ and $\Delta r_t^h \in \mathbb{R}^3$ describe translation and axis-angle rotation relative to time t , $v_t^h, \omega_t^h \in \mathbb{R}^3$ are future linear and angular velocities, and q_t^h is an activity logit. The prediction e_t^h is projected back into the hidden space when computing e_t^{h+1} .

D. Conditioning the Action Expert

The effect predictor retains a horizon-specific hidden context c_t^h . We combine this context with the corresponding physical prediction and project the result to the PI0.5 embedding dimension:

$$\tau_t^h = f_{\text{token}}([c_t^h; e_t^h]). \quad (5)$$

The four effect tokens are appended to the original prefix, $\tilde{X}_t = [X_t; \tau_t^1; \dots; \tau_t^4]$, and are visible to the flow-matching action expert. The original prefix is never replaced. Consequently, the policy can use the predicted physical future while retaining the full visual and linguistic context of the backbone.

E. Training Objective

For each valid horizon, simulator replay supplies a target effect $\hat{e}_t^h$ and mask M_t^h . The loss is

$$\mathcal{L} = \mathcal{L}_{\text{action}} + \lambda_E (\mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{act}}), \quad (6)$$

where $\mathcal{L}_{\text{reg}}$ is masked Smooth- L_1 loss over the 12 continuous values, $\mathcal{L}_{\text{act}}$ is binary cross-entropy over activity, and $\lambda_E = 1$. We train the effect chain without teacher forcing: the action expert is conditioned on model predictions, never on future labels.

TODO: Expand Eq. (6) to specify horizon aggregation, mask normalization, active/inactive weighting, and the exact action-loss definition implemented in the training code.

F. Replay-Based Effect Supervision

We construct effect labels by restoring the flattened simulator state stored at each demonstration frame and forwarding the simulator without re-executing the recorded action. For the supervised target, we measure pose displacement and future velocity at each offset in Eq. (2). Horizons that cross an episode boundary are masked. The conversion preserves the original RGB observations, robot state, action, instruction, task index, episode index, and frame index. On LIBERO-Object, the target occupies the same entity slot across all ten tasks. The task index is used only to select a training label and is not embedded by the policy or provided during rollout.

G. Effect Interventions

We intervene on predicted effects while fixing the observation, language, checkpoint, diffusion noise, and denoising schedule. *Mask* removes all effect tokens. *Zero-12D* sets the continuous values in Eq. (4) to zero while preserving the horizon context and activity logit. *Reverse* reverses the order of the four effect values, and *shuffle* substitutes values from another sample while retaining the receiving sample’s hidden contexts. These interventions distinguish dependence on the added token pathway from dependence on predicted physical values and their temporal organization.

TODO: Add the main architecture figure showing the PI0.5 backbone, implicit target representation, four autoregressive effect steps, action conditioning, and the training-only simulator-supervision pathway.

IV. EXPERIMENTS

A. Benchmark and Data

We evaluate on LIBERO-Object, which contains ten language-conditioned tabletop manipulation tasks [2]. The training set contains 454 available demonstration episodes and 66,984 frames. Replay enrichment yields 253,423 valid target-horizon labels, of which 170,107 are active. We verify frame-level alignment between the original and enriched datasets and observe a maximum action-alignment error of 2.90×10^{-8} . Terminal horizons without a valid successor state are masked.

B. Training Setup

Both vanilla PI0.5 and ArTE-VLA are initialized from `lerobot/pi05_base` and fine-tuned for 30,000 updates using eight NVIDIA H20 GPUs, global batch size 64, bfloat16 precision, and seed 1000. We use the same optimizer, learning-rate schedule, image resolution, action chunk length of 16, and five executed actions per inference call. ArTE-VLA adds the target head, effect chain, effect-token projection, and the auxiliary loss in Eq. (6); the remaining training recipe is unchanged.

TODO: Add the exact optimizer, learning rate and schedule, weight decay, image resolution, augmentation, checkpoint-selection rule, wall-clock training cost, and trainable-parameter overhead for reproducibility.

C. Evaluation Protocol

For each task, we evaluate both policies from the same 50 demonstration-derived initial simulator states, resulting in 500 paired episodes per method. Rollouts use synchronous batches of ten environments, ten flow-matching inference steps, and a fixed time horizon. Success is recorded without terminating the episode early. We report per-task and aggregate success, paired rescue and regression counts, and the exact two-sided McNemar test. Because these resets reconstruct expert demonstration states rather than the benchmark’s standard initial-state set, we keep this protocol separate from comparisons to previously published LIBERO results.

D. Baselines and Ablations

The primary baseline is recipe-matched vanilla PI0.5. To distinguish future prediction from auxiliary regularization and added capacity, the complete study also considers an effect predictor that does not condition actions, learned context tokens without structured values, current-target-state conditioning, non-autoregressive horizon prediction, and a privileged ground-truth-effect oracle. We evaluate removal, zero-value, reverse-order, and cross-sample shuffle interventions on the same ArTE-VLA checkpoint. These comparisons isolate three questions: whether the additional pathway improves control, whether future structure matters beyond additional tokens, and whether the trained action expert uses the predicted values.

TODO: Run and report every baseline and ablation named above, including capacity-matched and value-free controls. Remove any comparison from the prose if it is not completed.

E. Metrics

Closed-loop success is the primary metric. For effect prediction, we measure translation average and final displacement error, rotation error, linear and angular velocity error, and activity precision and recall. We also report prediction improvement over a zero-motion baseline. Intervention effects are measured by the fraction of changed action chunks, maximum absolute action difference, and relative root-mean-square (RMS) distance. Efficiency is measured with trainable parameter count, inference latency, and peak accelerator memory.

TODO: Add the effect-prediction and efficiency results promised by this section, with per-horizon errors, a zero-motion comparison, parameter overhead, latency, and peak memory; otherwise remove unreported metrics.

V. RESULTS

A. Closed-Loop Performance

Table I compares ArTE-VLA with vanilla PI0.5 and reserves matched columns for the auxiliary-only and value-free-token baselines. Aggregated over all ten tasks, target-effect conditioning improves success from 70.8% to 83.6%, an absolute gain of 12.8 percentage points. Across the 500 paired episodes, ArTE-VLA produces 127 rescues and 63 regressions. The corresponding exact two-sided McNemar test yields $p = 3.99 \times 10^{-6}$.

TODO: Repeat the main comparison with additional training seeds and report uncertainty intervals. Add a separate standard-LIBERO-initialization evaluation; do not merge it with the expert-demonstration-reset protocol used in Table I.

TABLE I

LIBERO-OBJECT SUCCESS OVER 50 PAIRED EPISODES PER TASK. AVAILABLE RESULTS ARE FROM ONE TRAINING SEED. BLANK CELLS INDICATE MATCHED BASELINES AWAITING TRAINING AND EVALUATION.

Target object	PI0.5	Auxiliary-only	Value-Free Tokens	ArTE-VLA	Δ vs. PI0.5
Alphabet Soup	76%			74%	-2
Cream Cheese	76%			94%	+18
Salad Dressing	88%			82%	-6
BBQ Sauce	96%			84%	-12
Ketchup	98%			64%	-34
Tomato Sauce	74%			92%	+18
Butter	4%			94%	+90
Milk	96%			70%	-26
Chocolate Pudding	6%			84%	+78
Orange Juice	94%			98%	+4
Overall	70.8%			83.6%	+12.8

TABLE II

MATCHED COMPONENT ABLATIONS ON LIBERO-OBJECT. BLANK CELLS DENOTE EXPERIMENTS THAT HAVE NOT YET BEEN COMPLETED.

Variant	Success	Δ vs. PI0.5
Auxiliary-only		
Value-Free Tokens		
Current-target conditioning		
Non-autoregressive effects		
Full ArTE-VLA	83.6%	+12.8

TODO: Add a compact results visualization or paired rescue/regression summary so that the strongly heterogeneous per-task behavior is visible without relying only on aggregate success.

The aggregate improvement is dominated by two large rescues. On Butter and Chocolate Pudding, vanilla PI0.5 succeeds in only 4% and 6% of episodes, whereas ArTE-VLA reaches 94% and 84%, respectively. The gain is therefore not a uniform shift across tasks. Ketchup and Milk decrease by 34 and 26 points, and BBQ Sauce decreases by 12 points. This heterogeneity indicates that target effects can resolve severe failures of a direct policy, but can also introduce task-dependent interference.

B. Component Ablations

Table II isolates the contribution of action conditioning, structured effect values, present-versus-future target information, and autoregressive horizon modeling. Each learned variant will use the same initialization, training budget, and evaluation protocol as the full model.

TODO: Populate Table II with separately trained, recipe-matched checkpoints and multi-seed uncertainty intervals.

C. Action Dependence on Effect Values

We perform a fixed-noise action audit on 32 observations drawn equally from Butter and Chocolate Pudding. Repeating the unmodified forward computation gives zero numerical action difference. Zeroing the 12 continuous effect values changes the action chunk for all 32 samples. The median maximum absolute action difference is 0.00956,

TABLE III

FIXED-NOISE EFFECT INTERVENTIONS ON 32 OBSERVATIONS. ACTION METRICS ARE MEASURED RELATIVE TO THE UNMODIFIED COMPUTATION. BLANK CELLS ARE PENDING; CLOSED-LOOP SUCCESS REQUIRES FULL-TASK EVALUATION.

Setting	Changed (%)	Max abs.	Rel. RMS (%)	Success (%)
Unmodified repeat	0	0	0	
Mask				
Zero-12D	100	0.00956	0.943	
Reverse				
Shuffle				

and the median relative RMS distance is 0.943%. Reverse-order and cross-sample shuffle interventions also produce consistent action changes under the fixed-noise protocol.

These results establish computational dependence of the action expert on the predicted physical values. They do not establish that those values alone account for the closed-loop improvement: each effect token also contains a learned horizon context, and the zero-value median lies slightly below the prespecified 1% relative-RMS decision threshold. Closed-loop interventions over the full task distribution are therefore necessary to quantify the independent control value of the structured prediction.

TODO: Extend the value intervention beyond the current 32 observations and two tasks. Run all-task action audits and closed-loop mask, zero-value, reverse-order, and cross-sample-shuffle interventions before making an independent value-use claim.

D. Task-Level Behavior

The paired outcomes suggest two complementary failure modes. For tasks on which vanilla PI0.5 rarely succeeds, target-effect supervision may provide an additional object-motion signal that disambiguates action generation. Conversely, the regressions on Ketchup, Milk, and BBQ Sauce may arise from inaccurate target effects, an implicit target representation that entangles object and task context, or competition between effect and visual-language tokens inside the action expert. Separating these explanations requires

comparing effect error and action sensitivity on matched rescue and regression episodes.

TODO: Add the matched rescue/regression diagnostic proposed above, including effect-error correlations and representative qualitative rollouts; otherwise keep these failure explanations explicitly speculative.

VI. DISCUSSION

The current results support a narrow conclusion: adding an autoregressive target-effect pathway to PI0.5 improves aggregate success on LIBERO-Object and measurably changes the action computation. They do not yet show that predicting future physical values is uniformly beneficial across tasks or that explicit values are the sole source of improvement. In particular, learned horizon contexts provide an additional conditioning channel that must be controlled with parameter-matched and value-free baselines.

The method also relies on privileged target-effect labels during training. This supervision is compatible with simulation and instrumented robot datasets, but application to unconstrained real-world data would require state estimation or pseudo-labeling. In addition, the current target representation is implicit and does not expose a calibrated present-time object pose. An explicit target estimator could improve interpretability, but would introduce association and geometry errors that the present architecture avoids.

Finally, Table I reports a single training seed and an expert-demonstration reset protocol. Multi-seed replication, matched auxiliary and capacity controls, standard LIBERO initial states, and evaluation beyond LIBERO-Object are required to determine whether the observed gain is robust and generalizes beyond the current setting.

TODO: Add standard LIBERO initial-state results, held-out task suites, and cross-simulator evaluation, or narrow the final generalization claims to the current LIBERO-Object expert-reset setting.

VII. CONCLUSION

We presented ArTE-VLA, a VLA architecture that predicts a compact physical future for the language-referred target and uses that prediction to condition action generation. The method requires privileged object states only for training-time supervision and retains the original RGB-language interface at inference. On LIBERO-Object, ArTE-VLA substantially improves aggregate success over a matched PI0.5 baseline, while intervention experiments confirm that the action expert responds to the predicted values. The task-level regressions and the contribution of hidden effect context identify important limitations. More broadly, our results motivate structured future effects as an inspectable interface between visual-language understanding and continuous robot control.

TODO: Revise the conclusion after the outstanding replications, ablations, and interventions; ensure that no final claim exceeds the completed evidence.

TODO: Complete the bibliography audit: expand coverage, replace arXiv entries with archival publication data where available, and verify author lists, titles, years, and identifiers.

REFERENCES

- [1] Physical Intelligence et al., “ $\pi_{0.5}$: A vision-language-action model with open-world generalization,” arXiv preprint arXiv:2504.16054, 2025.
- [2] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023.
- [3] Q. Zhao et al., “CoT-VLA: Visual chain-of-thought reasoning for vision-language-action models,” arXiv preprint arXiv:2503.22020, 2025.
- [4] J. Cen et al., “WorldVLA: Towards autoregressive action world model,” arXiv preprint arXiv:2506.21539, 2025.
- [5] C.-L. Cheang et al., “GR-2: A generative video-language-action model with web-scale knowledge for robot manipulation,” arXiv preprint arXiv:2410.06158, 2024.
- [6] C. Wen, X. Lin, J. So, K. Chen, Q. Dou, Y. Gao, and P. Abbeel, “Any-point trajectory modeling for policy learning,” arXiv preprint arXiv:2401.00025, 2024.
- [7] H. Bharadhwaj, R. Mottaghi, A. Gupta, and S. Tulsiani, “Track2Act: Predicting point tracks from internet videos enables generalizable robot manipulation,” arXiv preprint arXiv:2405.01527, 2024.

TODO: Before submission, set `\draftnotesfalse`, confirm the anonymous author block for review, remove identifying metadata and comments, and rerun the PaperPlaza PDF compliance check.